

Lecture 13b: Dimensionality Reduction

PCA · t-SNE · UMAP

Outline

The Need to Compress

PCA via Variance Maximization

Scaling, Choosing k , Pitfalls

t-SNE: probability-matching in low-dim

UMAP: faster, often better than t-SNE

PCA vs t-SNE vs UMAP: when to use which